On finitariness of the Full-Ground Local State Monad

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1 Background

1.1 Full-Ground Local State

The monad for Full-Ground local state is as follows [Kam+17]:

$$\begin{aligned} \mathcal{T} \quad &: \quad [\mathbb{W}, \mathsf{Set}] \to [\mathbb{W}, \mathsf{Set}] \\ (\mathcal{T}X)w \quad &:= \quad \underbrace{\int_{w \rightarrowtail w' \in w/U} \mathbb{H}w' \to \overbrace{\int^{w' \rightarrowtail w'' \in w'/U} (XU)w'' \times \mathbb{H}w''}^{P(XU \times \mathbb{H})w'}}_{U_*(P(XU \times \mathbb{H})^{\mathbb{H}})w} \end{aligned}$$

The monad's action on objects can be described as taking the right Kan extension along the forgetful functor U of the functor $P(XU \times \mathbb{H})^{\mathbb{H}}$. In this case $P : [\mathbb{E}, \mathsf{Set}] \to [\mathbb{E}, \mathsf{Set}]$ is the so-called hiding monad and $\mathbb{H} : \mathbb{E} \to \mathsf{Set}$ represents heaps ($\mathbb{E}$ has objects worlds and morphisms as initialisation data).

$$\mathbb{E} \xrightarrow{\quad U \quad} \mathbb{W} \qquad\qquad [\mathbb{W}, \mathsf{Set}] \underset{U_*}{\overset{U^*}{\underset{\longleftarrow}{\overset{\longrightarrow}{\bot}}}} [\mathbb{E}, \mathsf{Set}] \circlearrowleft P$$

1.2 Finitary Monads

- A monad is said to be finitary if its underlying endofunctor preserves ω-filtered colimits.
- An ω-filtered colimit is a colimit over an ω-filtered category.
- An ω-filtered category is a category $\mathcal{C}$ that satisfies all of the following conditions:
  - There exists at least one object in $\mathcal{C}$.
  - For any two objects $C, D \in \mathcal{C}$, there exists an object E and two morphisms $f : C \to E$, $g : D \to E$.
  - For any two parallel morphisms $f, g : C \to D$, there exists a morphism $h : D \to E$ such that $h \circ f = h \circ g$.

2 Non-finitariness of FGLS Monad

Theorem 1. The Full-Ground Local State Monad is not finitary.

Proof. We will show that there exists a filtered category $\mathcal{C}$ and functor $F : \mathcal{C} \to [\mathbb{W}, \mathsf{Set}]$ such that

$$\mathcal{T}(\mathit{colimit}\, F) \ncong \mathit{colimit}\, (\mathcal{T}F)$$

in the category of cocones for the functor F.

Take $\mathcal{C} = \omega$, the posetal category where objects are natural numbers and morphisms are orderings $m \leq n$. This category is ω-filtered. Further define $F : \omega \to [\mathbb{W}, \mathsf{Set}]$ to be $Fn = \Delta_n$ that maps each natural number n to the constant functor mapping to the downset of n and $m \leq n$ are mapped to inclusions.

Notice that we can simplify *colimit* F:

$$(\mathit{colimit}\, F)w = \langle \biguplus_{n\in\mathbb{N}} n\downarrow \Big/ \sim, (i \in n\downarrow \mapsto q_n(i))_{n\in\mathbb{N}}\rangle \cong \langle \mathbb{N}, (i \in n\downarrow \mapsto i)_{n\in\mathbb{N}}\rangle$$

which extends to an isomorphism of cocones $\mathcal{T}(\mathit{colimit}\, F) \cong \langle \mathcal{T}(\Delta_{\mathbb{N}}), (\mathcal{T}(\Delta_n \hookrightarrow \Delta_{\mathbb{N}}))_{n\in\mathbb{N}}\rangle$.

We also know that:

$$\mathit{colimit}\,(\mathcal{T}F)w \cong \langle \biguplus_{n\in\mathbb{N}} \mathcal{T}\Delta_n w \Big/ \sim, (f \in \mathcal{T}\Delta_n w \mapsto q_n(f))_{n\in\mathbb{N}}\rangle$$

where the quotienting says that $\langle n, f \in \mathcal{T}(\Delta_n)w\rangle \sim \langle m, g \in \mathcal{T}(\Delta_m)w\rangle$ iff they are the same morphisms after we compose them with their corresponding inclusions to $\mathcal{T}(\Delta_{n+m})$.

The above observations boil down to showing that for some $w \in \mathbb{W}$:

$$\biguplus_{n\in\mathbb{N}} \mathcal{T}\Delta_n w \Big/ \sim \not\cong \mathcal{T}(\Delta_{\mathbb{N}})w$$

where the mapping is given by $q_n(f \in \mathcal{T}(\Delta_n)w) \mapsto \mathcal{T}(\Delta_n \hookrightarrow \Delta_{\mathbb{N}})_w(f)$. We show that this mapping is not surjective.

Counterexample W.l.g. assume we only have one sort $\phi(C) = 1 + \mathsf{ref}_C$ and let $w = a : C$ be a singleton world. We define $\mathit{count} : \mathcal{T}(\Delta_{\mathbb{N}})w$ such that there cannot be element in $\mathcal{T}(\Delta_n)w$ for some n that represents it after post-composition with inclusion.

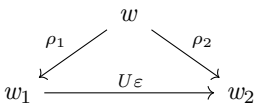
$$\begin{aligned}
&\mathit{count} : \int_{a:C\rightarrowtail w'\in a:C/U} \mathbb{H}w' \to \int^{w'\rightarrowtail w''\in w'/U} \mathbb{N}\times\mathbb{H}w'' \\
&\mathit{count}(\rho : a : C \rightarrowtail w')(h : \mathbb{H}w') = q_{1_{w'}}\langle n, h\rangle \\
\text{where}& \\
&go = \mathsf{fix}\,\lambda rec.\,\lambda\langle a \in w', S \subseteq w'\rangle.\,\mathbf{case}\,\pi_a(h) \\
&\qquad\qquad \mathbf{inl}\,().\,1 \\
&\qquad\qquad \mathbf{inr}\,b.\begin{cases} 1 & \text{if } b \in S \\ 1 + \mathit{rec}(b, \{b\}\cup S) & \text{otherwise}\end{cases} \\
&n = go\langle a, \{a\}\rangle
\end{aligned}$$

Intuitively, *count* traverses the heap h starting from a and counts the number of different memory cells that it has encountered. Notice that *count* is defined for every extension of $a : C$ to w' and so there cannot be a finite $n\downarrow$ that defines it and thus the mapping is not surjective. □

A note about the End in the FGLS monad The *count* function above is well-defined, i.e., it satisfies the End relation. To see why this is the case, notice that the End quantifies over all injections $\rho : w \to w'$, and thus it gives an ρ-ary product:

$$\{f_\rho : \mathbb{H}w' \to P(\Delta_{\mathbb{N}} \times \mathbb{H})w'\}_{\rho:w\to w'\in w/U}$$

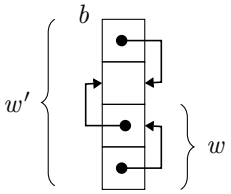
such that the following coherence condition holds:

For every initialisation of the form 

and heap $h : \mathbb{H}w_1$ we have:

$$P(\Delta_{\mathbb{N}} \times \mathbb{H})(\varepsilon)(f_{\rho_1}(h)) = f_{\rho_2}(\mathbb{H}\varepsilon h)$$

Intuitively, it says each component can only use the reachable part of a heap starting from locations in w. That is, take the following heap h where the location b is not reachable from locations in w.



The End relation says that the component f at world w' and heap h is completely determined by f at $w' \setminus \{b\}$ together with h without cell b. It is then clear to see that *count* satisfies the End relation as it traverses the heap *starting* from location a.

3 FGLS Monad on [Ctx,Set]

There is an inclusion functor $J : \mathbb{W} \hookrightarrow \mathsf{Ctx}$ which by precomposition extends to a functor $J^* : [\mathsf{Ctx}, \mathsf{Set}] \to [\mathbb{W}, \mathsf{Set}]$ and further an adjunction $[\mathsf{Ctx}, \mathsf{Set}] \underset{J_*}{\overset{J^*}{\underset{\longleftarrow}{\overset{\longrightarrow}{\bot}}}} [\mathbb{W}, \mathsf{Set}]$ by taking the right Kan extension along J. The idea behind J_* is that while locations l_1, l_2 in $\mathbb{W}$ represent distinct locations, variables a, b in Ctx represent non-distinct aliases to locations. To recover this aliasing, J_* considers all the possible ways this aliasing can happen.

The $J_* \vdash J_*$ adjunction gives us a way to extend the full-ground monad $\mathcal{T}$ to Ctx presheaves as:

$$X \mapsto J_*(\mathcal{T}(XJ)) : [\mathsf{Ctx}, \mathsf{Set}] \to [\mathsf{Ctx}, \mathsf{Set}]$$

We shall now give an explicit definition of this monad using the following theorem:

Theorem 2. Let $F : \mathcal{C} \to \mathcal{D}, A, B : \mathcal{C} \to \mathsf{Set}$. We will show that:

$$\mathrm{Ran}_F\, B^A(D) \cong \int_{f:D \to FC \in D/F} BC^{AC}$$

Proof.

$$\begin{aligned}
\mathrm{Ran}_F\, B^A(D) &\cong \int_{C \in \mathcal{C}} \mathcal{D}(D, FC) \to B^A C && \text{([Kel82] 4.24)} \\
&\cong \int_{f:D \to FC \in D/F} B^A C && \text{(Lemma 6)} \\
&\cong \int_{f:D \to FC \in D/F} BC^{AC} && \text{(Lemma 7)}
\end{aligned}$$

□

The Monad Notice that:

$$J_*(\mathcal{T}(XJ)) = J_* U_*(P(XJU)^{\mathbb{H}}) = \mathrm{Ran}_{JU}\,(P(XJU)^{\mathbb{H}})$$

We express the FGLS monad on $[\mathsf{Ctx}, \mathsf{Set}]$ using this observation and Theorem 2 as:

$$\begin{aligned}
\mathcal{T}_{\mathsf{C}} &: [\mathsf{Ctx}, \mathsf{Set}] \to [\mathsf{Ctx}, \mathsf{Set}] \\
\mathcal{T}_{\mathsf{C}} X &:= \int_{\rho:w \to w' \in w/JU} \mathbb{H}w' \to \int^{\rho':w' \to w'' \in w'/U} (XJU)(w'') \times \mathbb{H}w''
\end{aligned}$$

Theorem 3. The $\mathcal{T}_{\mathsf{C}}$ monad is non-finitary.

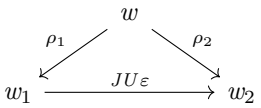
Proof. Follows directly from the non-finitary proof of $\mathcal{T}$. □

Naturality-style We can give an alternative description of $\mathcal{T}_{\mathsf{C}}$ using natural transformations by observing the action of the End. Intuitively, its action is that every f_{ρ_2} computation can be factored through a computation f_{ρ_1} that requires a smaller starting heap followed by an extension of the result in a compatible way. This forbids f from using locations that are not *reachable* from w while at the same time allowing it to be called in all compatible future worlds.

Recall that the End quantifies over all morphisms $\rho : w \to w'$, and thus gives an ρ-ary product:

$$\phi \in \prod_{w \xrightarrow{\rho} w' \in \mathsf{Obj}(w/JU)} \mathbb{H}w' \to P(XJU \times \mathbb{H})w'$$

such that the following coherence condition holds:

for every initialisation of the form:  and heap $h : \mathbb{H}w_1$ we have:

$$P(XJU \times \mathbb{H})(\varepsilon)(\phi_{\rho_1}(h)) = \phi_{\rho_2}(\mathbb{H}(\varepsilon)h)$$

If we compare this data with the data of the following natural tranformations we see that they are isomorphic:

$$\int_{\rho : w \to w' \in w/JU} \mathbb{H}w' \to P(XJU \times \mathbb{H})w' \cong [\mathbb{E}, \mathsf{Set}](\mathsf{Ctx}(w, JU(-)) \times \mathbb{H}, P(XJU \times \mathbb{H}))$$

We can go further and define the functorial action, unit and join of the FGLS monad in terms of this naturality style which will be used in §4.2 to define the finitary monad.

Functorial action

$$\begin{aligned} \mathcal{T}_{\mathsf{C}}X(f : w \to w') \ &: \ \mathcal{T}_{\mathsf{C}}Xw \to \mathcal{T}_{\mathsf{C}}Xw' \\ &\phi \mapsto \mathsf{Ctx}(w', JU(-)) \times \mathbb{H} \xrightarrow{(-\circ f)\times 1} \mathsf{Ctx}(w, JU(-)) \times \mathbb{H} \xrightarrow{\phi} P(XJU \times \mathbb{H}) \end{aligned}$$

Unit of Monad

$$\begin{aligned} \eta_X \ &: \ X \Rightarrow \mathcal{T}_{\mathsf{C}}X \\ x : Xw \mapsto \mathsf{Ctx}(w, JU(-)) \times \mathbb{H} &\xrightarrow{X(-)(x)\times 1} XJU \times \mathbb{H} \xrightarrow{\eta^P_{XJU\times\mathbb{H}}} P(XJU \times \mathbb{H}) \end{aligned}$$

where $X(-)(x) : \mathsf{Ctx}(w, JU(-)) \Rightarrow XJU$ is defined as $f \mapsto X(f)(x)$.

Join of Monad

$$\begin{aligned} \mu_X \ &: \ {\mathcal{T}_{\mathsf{C}}}^2X \to \mathcal{T}_{\mathsf{C}}X \\ \phi \mapsto \mathsf{Ctx}(w, JU(-)) \times \mathbb{H} &\xrightarrow{\phi} P(\mathcal{T}_{\mathsf{C}}XJU \times \mathbb{H}) \xrightarrow{P(\mathsf{eval})} P^2(XJU \times \mathbb{H}) \xrightarrow{\mu^P_{XJU\times\mathbb{H}}} P(XJU \times \mathbb{H}) \end{aligned}$$

where $\mathsf{eval} : (\mathcal{T}_{\mathsf{C}}X)JU \times \mathbb{H} \Rightarrow P(XJU \times \mathbb{H})$ is defined as $\langle \phi, h \rangle_w \mapsto \phi_w \langle 1_w, h \rangle$.

4 Becoming Finitary

A natural question to ask is how to carve out a subset of the function space described by the FGLS monad such that the monad it induces is finitary. Our goal is to prove a representation result between this finitary monad and the parametrised algebraic theory for FGLS. Since the PAT for FGLS has models in $[\mathsf{Ctx}, \mathsf{Set}]$, we consider the finitary version of the FGLS monad lifted to Ctx presheaves (see §3).

4.1 Preliminaries

We first introduce some properties and constructions that are used to describe the finitary monad.

4.1.1 Heaplet Initialisation Category

The heaplet initialisation category $\underline{\mathbb{E}}$ is defined as:

- Objects: Sort-preserving injective renamings between worlds $w^- \rightarrowtail w^+$.
- Morphisms: Injective sort-preserving ε^-, sort-preserving ε^+, and a $h_\varepsilon : \underline{\mathbb{H}}(w_2^- \ominus \varepsilon^-, w_2^+)$ such that:

$$\begin{array}{ccc} w_1^+ & \xrightarrow{\varepsilon^+} & w_2^+ \\ {\scriptstyle\rho}\uparrow & & \uparrow{\scriptstyle\sigma} \\ w_1^- & \xrightarrow[\varepsilon^-]{} & w_2^- \end{array}$$

We have functors $U_1 : \underline{\mathbb{E}} \to \mathbb{W}$, $U_2 : \underline{\mathbb{E}} \to \mathsf{Ctx}$ that project the first and second world correspondingly, $\underline{J} : \mathbb{E} \hookrightarrow \underline{\mathbb{E}}$ which maps w to the corresponding identity morphism, and the heaplet functor can be defined as $\underline{\mathbb{H}} : \underline{\mathbb{E}} \to \mathsf{Set}$ (notice that we consider the heaplet functor to be of the form $\mathbb{W}^{\mathsf{op}} \times \mathsf{Ctx} \to \mathsf{Set}$).

An order on morphisms We say that a morphism ε_2 is *at least as concrete* as ε_2 (notation $\varepsilon_1 \leq \varepsilon_2$), when $\varepsilon_1, \varepsilon_2$ have the same domain:

$$\begin{array}{ccccc} w_2^+ & \xleftarrow{U_2\varepsilon_2} & w^+ & \xrightarrow{U_2\varepsilon_1} & w_1^+ \\ \uparrow & & \uparrow{\scriptstyle \rho} & & \uparrow \\ w_2^- & \xleftarrow[U_1\varepsilon_2]{} & w^- & \xrightarrow[U_1\varepsilon_1]{} & w_1^- \end{array}$$

and satisfy:

$$\text{every } x \in \mathsf{Img}_{U_2\varepsilon_2}[\mathsf{Fibers}_{U_2\varepsilon_1}[w_1^+]] \text{ is a singleton set}$$

The idea is that ε_2 does not relate more dangling pointers from $w^+ \ominus \rho$ than ε_1.

Example 1.

Lemma 4. $\leq$ forms a preorder

Extension Functor We define the extension functor as a means to add w'-many dangling pointers.

$$\begin{aligned} (- + w') &: \underline{\mathbb{E}} \to \underline{\mathbb{E}} \\ ((w^- \xrightarrow{\rho} w^+) + w') &= w^- \xrightarrow{\rho} w^+ \xhookrightarrow{inl} w^+ \oplus w' \\ (\varepsilon + w') &= \langle U_1\varepsilon,\, U_2\varepsilon \oplus 1_{w'},\, \underline{\mathbb{H}}\langle 1, inl\rangle(\eta_\varepsilon)\rangle \end{aligned}$$

Further, for every functor $G : \underline{\mathbb{E}} \to \mathsf{Set}$, we have:

$$\begin{aligned} \mathsf{ext}_G^{w'} &: G \Rightarrow G(- + w') \\ \mathsf{ext}_G^{w'}(w^- \xrightarrow{\rho} w^+) &= G\langle 1_{w^-},\, inl^{w^+ \xhookrightarrow{inl} w^+ \oplus w'},\, \emptyset\rangle \end{aligned}$$

We will often drop G from $\mathsf{ext}_G^{w'}$ when it is clear from the context.

4.1.2 Templates

We introduce the notion of a *template* that represents a collection of objects in $\underline{\mathbb{E}}$. Specifically, a template is a set I seen as a discrete category, together with a functor $T : I \to \underline{\mathbb{E}}$. We say that a template is finite when I is a finite set.

We would like to represent all the possible heap extensions that can be derived from a template. For a template T, its heap extensions are given by the following functor:

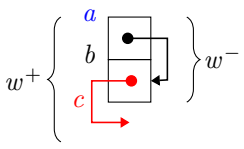
$$\begin{aligned} &F^* : \mathbb{E} \to \mathsf{Set} \\ &F^* = \mathit{colimit}\,(I \xrightarrow{T} \underline{\mathbb{E}} \xrightarrow{y} [\underline{\mathbb{E}}, \mathsf{Set}] \xrightarrow{(-)J} [\mathbb{E}, \mathsf{Set}]) \cong \coprod_{i \in I} \underline{\mathbb{E}}(T(i), \underline{J}(-)) \end{aligned}$$

4.1.3 Stencils

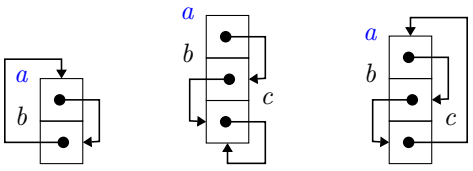
Stencils are used to represent heaps with a common segment. It is parametrised over a world w that represents the public locations, and it is defined using heaplets with dangling pointers. Dangling pointers represent the points by which the stencil can be extended into a bigger heaplet/heap.

$$\begin{aligned} \mathcal{S} &: \mathsf{Ctx}^{\mathsf{op}} \to [\underline{\mathbb{E}}, \mathsf{Set}] \\ \mathcal{S}_w &= \mathsf{Ctx}(w, U_2(-)) \times \underline{\mathbb{H}} \\ \mathcal{S}_f\langle g, h\rangle &= \langle g \circ f, h\rangle \end{aligned}$$

Example 2. Consider the following stencil given diagrammatically and assume all locations are of a single sort C (locations in blue are directly accessible from the public world $w = a$, pointers in red are dangling):



This stencil represents the heaps that have this segment as starting point from a, to draw a few:



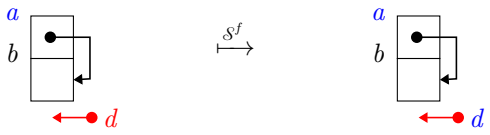
Rebasing a Stencil We define a covariant action of morphisms Ctx acting on a stencil which we call *rebasing*. For every morphism $f : w \to w' \in \mathsf{Ctx}$:

$$\mathcal{S}^f \ : \ \mathcal{S}_w(- + w' \ominus f) \Rightarrow \mathcal{S}_{w'}(- + w' \ominus f)$$
$$\mathcal{S}^f_{w^- \xrightarrow{\rho} w^+} \langle g^{w \to w^+ \oplus w' \ominus f}, h \rangle \mapsto \langle w' \xrightarrow{\cong} w \oplus w' \ominus f \xrightarrow{[g, inr]} w^+ \oplus w' \ominus f, h \rangle$$

Lemma 5. The functorial action $\mathcal{S}_f$ is the retraction of the rebasing operation, that is:

$$\mathcal{S}_{f\,(- + w' \ominus f)} \circ \mathcal{S}^f = 1$$

Example 3. Consider the following extension $w = a : C, w' = a : C, d : C'$ and $f = inl$. We have that $w' \ominus f = d : C'$. Consider a stencil that has d as dangling reference:



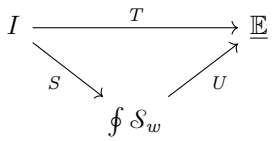
The rebasing operation simply makes the location d publicly accessible from w'.

4.1.4 Stencil Cover

We view stencils accessible via a world w as a category by considering the category of elements for functor $\mathcal{S}_w$ i.e., $\oint \mathcal{S}_w$, defined as:

- Objects: $\langle \rho : w^- \rightarrowtail w^+, s \in \mathcal{S}_w(\rho) \rangle$
- Morphisms: $\langle \rho, s \rangle \xrightarrow{\varepsilon} \langle \sigma, s' \rangle$ is a morphism $\varepsilon : \rho \to \sigma \in \underline{\mathbb{E}}$, such that $\mathcal{S}_w(\varepsilon)(s) = s'$

A *stencil cover* is a collection of stencils that is compatible with a particular template. That is, a w-accessible T-cover, is a functor $S : I \to \oint \mathcal{S}_w$ such that:



We have a natural transformation $\mathsf{cov} : T^* \Rightarrow \mathsf{Ctx}(w, JU(-)) \times \mathbb{H}$ given by:

$$\langle i, \varepsilon^{T(i) \to \sigma} \rangle \mapsto \mathcal{S}_w(\varepsilon)(\pi_2(Si))$$

We say that a cover is *complete* when cov is an isomorphism.

4.1.5 Heaplet Hiding Monad

We further generalise the hiding monad to heaplets as follows (see §A.1 for the functorial action):

$$\underline{P} : \ [\underline{\mathbb{E}}, \mathsf{Set}] \to [\underline{\mathbb{E}}, \mathsf{Set}]$$
$$(\underline{P}A)(\rho : w^- \rightarrowtail w^+) = \int^{w^- \rightarrowtail w' \in w^-/U} A(inl^{w' \hookrightarrow w' \oplus w^+ \ominus \rho})$$

which takes the coend of $(w^-/U)^{\mathsf{op}} \times w^-/U \xrightarrow{\pi_2} w^-/U \xrightarrow{U} \mathbb{E} \xrightarrow{J} \underline{\mathbb{E}} \xrightarrow{(- + w^+ \ominus \rho)} \underline{\mathbb{E}} \xrightarrow{A} \mathsf{Set}$. We have a natural isomorphism $\underline{P}(X \odot \underline{\mathbb{H}})\underline{J} \cong P(X \odot \mathbb{H})$ between the hiding monad on heaplets with the one on heaps when $A = X \odot \underline{\mathbb{H}}$.

We consider the action of $\underline{P}A$ on morphisms as consisting of two parts:

- An action between elements of the equivalence classes (does not satisfy functoriality):

$$\underline{\mathbb{P}}A(\varepsilon : \rho \to \sigma) : \coprod_{w_1^- \rightarrowtail w'} A(inl^{w' \to w' \oplus (w_1^+ \ominus \rho)}) \to \coprod_{w_2^- \rightarrowtail w'} A(inl^{w' \to w' \oplus (w_2^+ \ominus \sigma)})$$

- A quotienting step given by the coend to make it functorial:

$$\underline{P}A(\varepsilon)(q_{\rho'}\, a) = [\underline{\mathbb{P}}A(\varepsilon)(inj_{\rho'}\, a)]$$

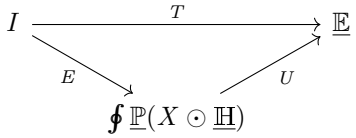
We further consider a variation of the category of elements for $\underline{P}A$ that will circumvent the need for representatives in later constructions. Define the category of elements $\oint \underline{\mathbb{P}}A$, as follows:

- Objects: $\langle \rho \in \underline{\mathbb{E}}, p \in \underline{\mathbb{P}}A(\rho) \rangle$
- Morphisms: $\langle \rho, p \rangle \to \langle \sigma, s \rangle$ is a morphism $\varepsilon : \rho \to \sigma \in \underline{\mathbb{E}}$ such that $\underline{P}A(\varepsilon)[p] = [s]$

Thus the objects of this category are the elements of the equivalence classes of $\underline{P}A$ and not the equivalence classes themselves.

4.1.6 Evaluations

We associate a template with its evaluations using the above hiding monad as the codomain. Define an X-valued T-evaluation as a functor $E : I \to \oint \underline{\mathbb{P}}(X \odot \underline{\mathbb{H}})$ that is compatible with T:



We have a natural transformation $\mathsf{ev} : T^* \Rightarrow P(X \odot \mathbb{H})$ given by:

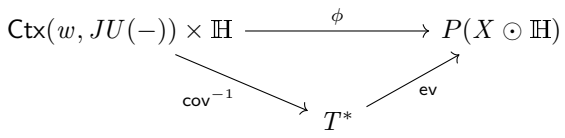
$$\langle i, \varepsilon^{T(i) \to \sigma} \rangle \mapsto \underline{\cong}^{\underline{P}(X \odot \underline{\mathbb{H}})\underline{J}, P(X \odot \mathbb{H})} (\underline{P}(X \odot \underline{\mathbb{H}})(\varepsilon)[\pi_2(Ei)]$$

4.2 Finitary Monad

A $\phi \in (\mathcal{T}_{\mathsf{C}} X)w$ is finitary when it is generated by its actions on a finite number of *heaplets*.

$$\begin{aligned} \mathcal{T}_{\mathsf{C}}^{\mathsf{fin}} :\ & [\mathsf{Ctx}, \mathsf{Set}] \to [\mathsf{Ctx}, \mathsf{Set}] \\ (\mathcal{T}_{\mathsf{C}}^{\mathsf{fin}} X)w = \{\phi \in (\mathcal{T}_{\mathsf{C}} X)w : & \textit{exists a finite template } T, \\ & \textit{a complete } w\textit{-accessible } T\textit{-cover } S, \\ & \textit{an } X\textit{-valued } T\textit{-evaluation } E, \\ & \textit{such that } \mathsf{ev}_E \circ \mathsf{cov}_S^{-1} = \phi\} \end{aligned}$$

- We consider $\phi \in (\mathcal{T}_{\mathsf{C}} X)w$ using its naturality description (see §3).
- We require a finite template, and thus we have a finite number of stencils.
- We require a complete cover, so $\mathsf{cov} : F^* \Rightarrow \mathsf{Ctx}(w, U(-)) \times \mathbb{H}$ is an isomorphism. This gives us that every $\langle f, h \rangle$ is mapped to exactly one stencil Si and morphism ε such that $\mathcal{S}_w(\varepsilon)(\pi_2(Si)) = \langle f, h \rangle$.
- We require that morphism ϕ lifts through cov^{-1} using the morphism ev induced by the evaluation E. This means that the following diagram commutes:



4.2.1 Action on Morphisms

We show that the factorial action of the $\mathcal{T}_{\mathsf{C}}^{\mathsf{fin}}$ monad on morphisms lifts through $\mathcal{T}_{\mathsf{C}}$, that is, for every $f : w \to w'$ we have functor $G : I' \to y/\mathcal{S}_{w'}$ and morphism h such that:

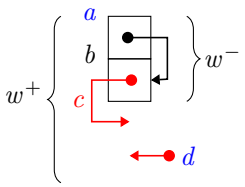
$$\begin{array}{ccccc} \mathsf{Ctx}(w', JU-) & \xrightarrow{(-\circ f)\times 1} & \mathsf{Ctx}(w, JU-) & \xrightarrow{\phi} & P(XJU \times \mathbb{H}) \\ {\scriptstyle \mathsf{cov}\underline{J}(-)}\uparrow\downarrow\cong & & {\scriptstyle \mathsf{cov}\underline{J}(-)}\uparrow\downarrow\cong & \nearrow_{\mathsf{ev}\underline{J}(-)} & \\ G^*\underline{J} & & \xrightarrow[h_{\underline{J}(-)}]{} & F^*\underline{J} & \end{array}$$

Proof. We utilize the structure of the functorial action on morphisms, that is, for a morphism $\varepsilon : \rho \to \sigma$, $inj_{\rho'}\, a$, $\underline{P}A(\varepsilon)[inj_{\rho'}\, a]$ will give a

Every morphism $f : w \to w' \in \mathsf{Ctx}$ induces a functor $\oint \mathcal{S}^f : \oint \mathcal{S}_w \to \oint \mathcal{S}_{w'}$ which extends a stencil with $w' \ominus f$ pointers and then rebases it:

$$\begin{aligned} \oint \mathcal{S}^f \ &: \ \oint \mathcal{S}_w \to \oint \mathcal{S}_{w'} \\ \langle \rho, s\rangle &\mapsto \langle \rho + (w' \ominus f), \mathcal{S}^f(\mathsf{ext}^{w'\ominus f}(s))\rangle \\ \varepsilon &\mapsto \varepsilon + (w' \ominus f) \end{aligned}$$

Example 4. Consider the stencil from Example 2 and the extension of $w = a : C$ to $w' = a : C, d : C'$. The resulting stencil has an extra dangling pointer $d : C'$ and so any extension of the stencil must be a heap that is defined with a location of sort C'.



Consider $G = \oint \mathcal{S}^f F : I \to y/\mathcal{S}_{w'}$ and $h = \coprod y\langle 1, inl\rangle$.

$$\begin{array}{ccccc} \mathsf{Ctx}(w', JU-) & \xrightarrow{(-\circ f)\times 1} & \mathsf{Ctx}(w, JU-) & \xrightarrow{\phi} & P(XJU \times \mathbb{H}) \\ {\scriptstyle \mathsf{cov}\underline{J}(-)}\uparrow\downarrow\cong & & {\scriptstyle \mathsf{act}^{\phi}_{\underline{J}(-)}}\uparrow\downarrow\cong & \nearrow_{\mathsf{res}^{\phi}_{\underline{J}(-)}} & \\ (\oint \mathcal{S}^f F)^*\underline{J} & & \xrightarrow[\coprod y\langle 1,inl\rangle_{\underline{J}(-)}]{} & F^*\underline{J} & \end{array}$$

That $\mathsf{cov}\underline{J}(-)$ is iso and that the square commutes follows from Lemma 5 by observing that:

$$\begin{array}{ccc} \mathsf{Ctx}(w', JU-) & \xrightarrow{(-\circ f)\times 1} & \mathsf{Ctx}(w, JU-) \\ \uparrow\downarrow\cong & & \uparrow\downarrow\cong \\ \mathcal{S}_{w'}\underline{J} & \xrightarrow{\mathcal{S}_f} & \mathcal{S}_w\underline{J} \end{array}$$

Orpheas: TODO: Latexify the handwritten proof. Further, every $f : w \to w'$ induces a natural transformation $\coprod y\langle 1, inl\rangle : (\oint \mathcal{S}^f \circ F)^* \Rightarrow F^*$:

$$\langle i, \varepsilon^{(UF(i)+w'\ominus f)\to\sigma}\rangle \mapsto \langle i, \varepsilon \circ \langle 1, inl, \emptyset\rangle\rangle$$

□

4.2.2 Unit of Monad

We show that the unit of the $\mathcal{T}_{\mathsf{C}}^{\mathsf{fin}}$ monad lifts through $\mathcal{T}_{\mathsf{C}}$, that is, $\eta_{X,w}(x)$ satisfies:

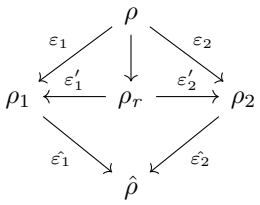
$$\begin{array}{ccc} \mathsf{Ctx}(w, JU(-)) \times \mathbb{H} & \xrightarrow{\eta^P \circ X(-)(x)\times 1} & P(XJU \times \mathbb{H}) \\ {\scriptstyle \mathsf{cov}\underline{J}(-)}\uparrow\downarrow\cong & \nearrow_{\mathsf{ev}\underline{J}(-)} & \\ F^*\underline{J} & & \end{array}$$

Proof. Consider the stencil $s = \langle 1_w, \emptyset \rangle : \mathcal{S}_w(\emptyset \rightarrowtail w)$ that has only w dangling pointers, and let $F(*) = s : 1 \to y/\mathcal{S}_w$. That $\mathsf{cov}\underline{J}(-)$ is iso is given by $\langle f, h \rangle \mapsto inj_* \langle ? : \emptyset \rightarrowtail w, f, h \rangle$ and ev is defined as:

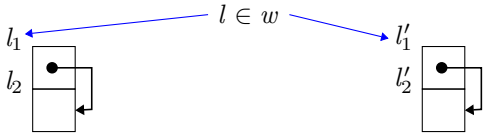
$$\mathsf{ev}(inj_* \varepsilon) = \underline{P}(XU_2 \times \underline{\mathbb{H}})(\varepsilon)(\eta^{P}_{\emptyset \to w} \langle x, \emptyset \rangle)$$

□

4.2.3 Kleisli Extension of Monad

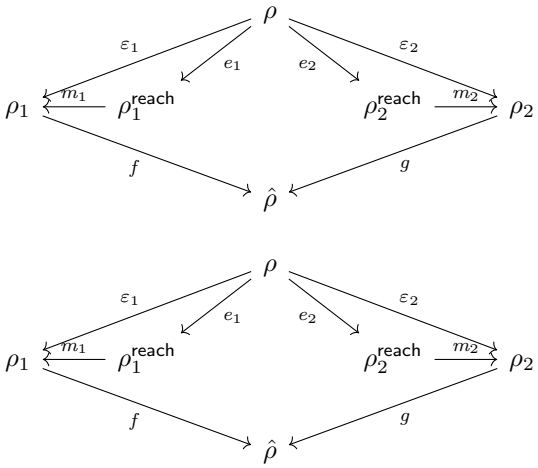


$\varepsilon_1 \oplus_\rho \varepsilon_2 :=$ Treat pointers from ρ_1, ρ_2 as distinct unless they come from the same domain and do this recursively by following the allocated data from ε_1, ε_2.



Reachable/Unreachable Factorisation System The category $\underline{\mathbb{E}}$ has an orthogonal factorisation system (E, M) that splits a morphism $\varepsilon : \rho_1 \to \rho_2$ into an $e : \rho_1 \to \rho_1^{\mathsf{reach}}$ representing the reachable part of η_ε and $m : \rho_1^{\mathsf{reach}} \to \rho_2$ representing the rest unreachable part.

We can then define local independent morphisms f, g as:



such that if $U^+(f)(a) = U^+(g)(b)$ then $a \in \mathsf{Im}(U^+(m_1))$ and $b \in \mathsf{Im}(U^+(m_2))$.

In words: if two pointers are related then they must come from the reachable part.

A Appendix

A.1 Hiding Monad on Heaplets

Recall the action of $\underline{P}$ on objects:

$$\underline{P} : \ [\underline{\mathbb{E}}, \mathsf{Set}] \to [\underline{\mathbb{E}}, \mathsf{Set}]$$
$$(\underline{P}A)(\rho : w^- \rightarrowtail w^+) = \int^{w^- \rightarrowtail w' \in w^- / U} A(i^{w' \hookrightarrow w' \oplus w^+ \ominus \rho})$$

The action on morphism $\varepsilon = \begin{array}{ccc} w_1^+ & \xrightarrow{U_2\varepsilon} & w_2^+ \\ {\scriptstyle\rho}\uparrow & & \uparrow{\scriptstyle\sigma} \\ w_1^- & \xrightarrow[U_1\varepsilon]{} & w_2^- \end{array}$, $h_\varepsilon : \underline{\mathbb{H}}(w_2^- \ominus U_1\varepsilon, w_2^+)$ is given as:

$$\begin{array}{l} \underline{P}A(\varepsilon) : \ \underline{P}A\rho \to \underline{P}A\sigma \\ q_{w_1^- \xrightarrow{\rho'} w'}\, a \mapsto q_{w_2^- \xrightarrow{(U_1\varepsilon)^*\rho'} \rho' \oplus_{w_1^-} U_1\varepsilon}\, A(\varepsilon')(a) \end{array}$$

Where $(U_1\varepsilon)^*\rho'$ comes from: $\begin{array}{ccc} w_1^- & \xrightarrow{U_1\varepsilon} & w_2^- \\ {\scriptstyle\rho'}\downarrow & & \downarrow{\scriptstyle(U_1\varepsilon)^*\rho'} \\ w' & \xrightarrow[(\rho')^* U_1\varepsilon]{} & \rho' \oplus_{w_1^-} U_1\varepsilon \end{array}$

and ε' is: $\begin{array}{ccc} w' \oplus (w_1^+ \ominus \rho) & \xrightarrow{[(\rho')^* U_1\varepsilon, \psi \circ U_2\varepsilon \circ \rho^c]} & \rho' \oplus_{w_1^-} U_1\varepsilon \oplus w_2^+ \ominus \sigma \\ \uparrow & & \uparrow \\ w' & \xrightarrow[(\rho')^* U_1\varepsilon]{} & \rho' \oplus_{w_1^-} U_1\varepsilon \end{array}$, $\underline{\mathbb{H}}(\cong, \psi)(\eta_\varepsilon)$

where ψ is given by:

$$w_2^+ \xrightarrow{\cong} w_1^- \oplus (w_2^- \ominus U_1\varepsilon) \oplus (w_2^+ \ominus \sigma) \hookrightarrow \rho' \oplus_{w_1^-} U_1\varepsilon \oplus w_2^+ \ominus \sigma$$

We will often consider the action of $\underline{P}A(\varepsilon)$ as two parts:

- An action between elements of the equivalence classes (does not satisfy functoriality):

$$\begin{array}{l} \underline{\mathbb{P}}A(\varepsilon : \rho \to \sigma) \ : \coprod_{w_1^- \rightarrowtail w'} A(inl^{w' \to w' \oplus (w_1^+ \ominus \rho)}) \to \coprod_{w_2^- \rightarrowtail w'} A(inl^{w' \to w' \oplus (w_2^+ \ominus \sigma)}) \\ \underline{\mathbb{P}}A(\varepsilon)(inj_{\rho'}\, a) = inj_{U_1\varepsilon * \rho'} A(\varepsilon')(a) \end{array}$$

- A quotienting step given by the coend to make it functorial:

$$\underline{P}A(\varepsilon)(q_{\rho'}\, a) = [\underline{\mathbb{P}}A(\varepsilon)(inj_{\rho'}\, a)]$$

A.2 End related lemmas

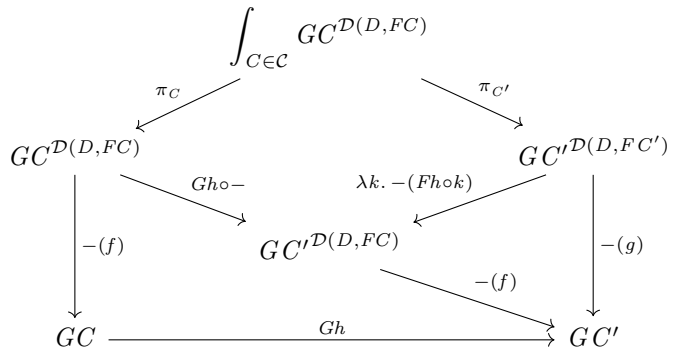
Unless otherwise specified let $A, B, G : \mathcal{C} \to \mathsf{Set}$, $F : \mathcal{C} \to \mathcal{D}$ and the following morphisms in D/F:

$$\begin{array}{ccc} FC & \xrightarrow{Fh} & FC' \\ & {\scriptstyle f}\nwarrow \quad \nearrow{\scriptstyle g} & \\ & D & \end{array}$$

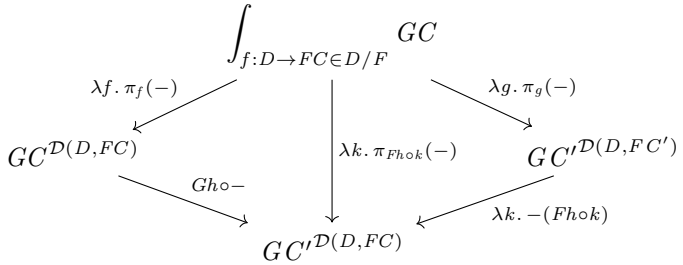
Lemma 6.

$$\int_{C \in \mathcal{C}} GC^{\mathcal{D}(D, FC)} \cong \int_{f : D \to FC \in D/F} GC$$

Proof. The morphism $\int_{C \in \mathcal{C}} GC^{\mathcal{D}(D,FC)} \dashrightarrow \int_{f : D \to FC \in D/F} GC$ is from the following wedge:



The morphism $\int_{f:D\to FC\in D/F} GC \dashrightarrow \int_{C\in\mathcal{C}} GC^{\mathcal{D}(D,FC)}$ is from the following wedge:

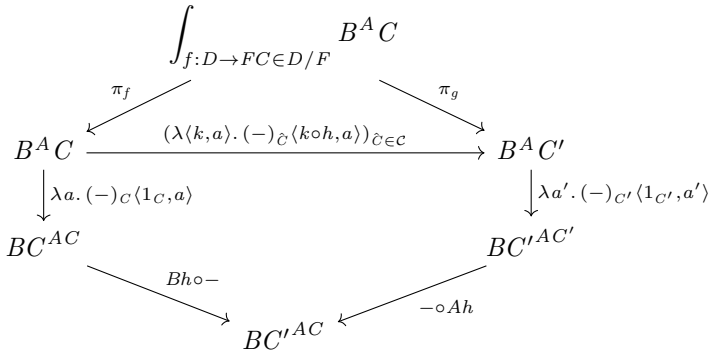


It follows directly that these two induced morphisms are inverses, and thus the two ends are isomorphic. □

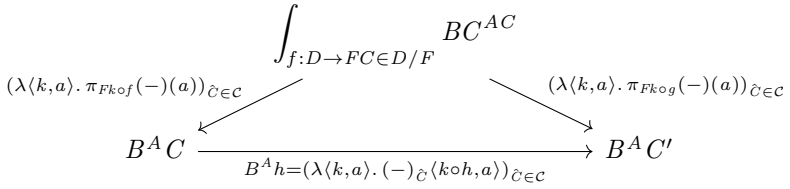
Lemma 7.

$$\int_{f:D\to FC\in D/F} B^{A}C \cong \int_{f:D\to FC\in D/F} BC^{AC}$$

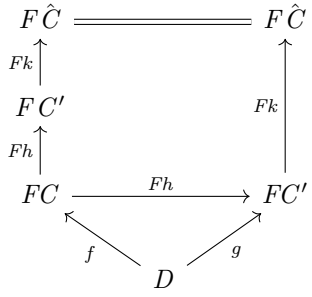
Proof. The morphism $\int_{f:D\to FC\in D/F} B^{A}C \dashrightarrow \int_{f:D\to FC\in D/F} BC^{AC}$ is from the following wedge:



The morphism $\int_{f:D\to FC\in D/F} BC^{AC} \dashrightarrow \int_{f:D\to FC\in D/F} B^{A}C$ is from the following wedge:



which commutes by:



□

That the end projections form a natural transformation and that the two induced morphisms are inverses is not hard to prove.

References

[Kel82] G. M. Kelly. *Basic Concepts of Enriched Category Theory*. Cambridge University Press, 1982.

[Kam+17] Ohad Kammar, Paul Blain Levy, Sean K. Moss, and Sam Staton. "A monad for full ground reference cells". In: *LICS*. 2017. URL: https://doi.org/10.1109/LICS.2017.8005109.